Paper: An Analysis of Facebook Photo Caching

Sample

1. The impact of cache size and choice of cache eviction algorithm.
2. Design tradeoffs of collaborative cache.
3. What are the key challenges of data collection for evaluating a system of this scale? What can introduce bias and inaccuracies?

Boyan Fan

1. What does a multi-layer cache actually do to the workload? (workload transformed by layers)
2. Why do we need multiple layers in the first place? (different localities)
3. Once the workload has been transformed, what should each cache keep? (cache eviction algorithms)

Igor Tiapkin

1. Where should image resizing happen: client, Edge, or near Origin? What are the tradeoffs?
2. Should Edge routing consider cache state or client-to-Edge stickiness in addition to latency, load, and peering cost?
3. How much should Edge Caches collaborate, and at what geographic scale?
4. How stable are the paper's results across different photoId samples?

Zihan Wang

1. How reliable is the browser cache hit rate inferred from missing Edge requests, given possible CDN routing and logging errors?
2. Does sampling by photo ID preserve the same cache competition and eviction behavior as the full workload?
3. Which conclusions from the 2013 photo workload still generalize to today's recommendation-driven, video-heavy platforms?

Chuchu Zeng

1. What are the main optimization goals for a large-scale caching system: object hit ratio, byte-hit ratio, latency, or backend I/O reduction?
2. How do viral photos and repeatedly accessed photos differ in their caching behavior?
3. How do failures and remote retries influence the design of the caching and storage hierarchy?

Sukhman Sra

1. What are the caching tradeoffs of storing multiple resized versions of the same photo as separate objects?
2. Why does browser cache effectiveness vary so much with user activity level?
3. Why does Facebook treat the Origin Cache as a single logical cache across multiple data centers despite the potential increase in latency?